WEEK-1
ADVANCE WEB PROGRAMMING

What is HTML?

HTML (HyperText Markup Language) is the **skeleton of a web page**. It defines **structure and content**.
Web browsers interpret HTML to display text, images, links, forms, and other media.

Example:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Welcome to AWP</title>
</head>
<body>
  <h1>Advance Web Programming</h1>
  <p>This course will make you a modern web developer!</p>
  <a href="https://developer.mozilla.org">Learn More</a>
</body>
</html>
```

Explanation:

- `<html>` is the root element of the page.
- `<head>` contains metadata (title, charset, links).
- `<body>` contains all visible content.
- `<h1>` defines the main heading, and `<p>` defines a paragraph.
- `<a>` creates a hyperlink.

What is CSS?

CSS (Cascading Style Sheets) controls the **presentation** of a web page — colors, fonts, spacing, layout, and responsiveness.

It allows us to separate **design** from **structure**.
For example, HTML defines *what* is on the page, and CSS defines *how* it looks.

Example:

```
body {
```

```
  background-color: #121212;
  color: white;
  font-family: 'Poppins', sans-serif;
}

h1 {
  color: #00bcd4;
  text-align: center;
}

a {
  color: yellow;
  text-decoration: none;
}
a:hover {
  color: orange;
}
```

Explanation:

- The `body` selector changes the background and text color.
- The `h1` selector styles the heading and centers it.
- The `a:hover` pseudo-class changes the color when the user hovers the mouse.

What is JavaScript?

JavaScript (JS) adds **behavior and interactivity** to web pages.
It allows dynamic updates, animations, form validation, API calls, and much more.
Modern JavaScript (ES6 and beyond) introduces new syntax and features for writing **cleaner, shorter, and more powerful code**.

Variables: var, let, const

JavaScript variables are used to store data values.

Keyword	Scope	Re-declare	Reassign	Hoisting	Common Use
`var`	Function	✓	✓	✓	Legacy code
`let`	Block	✗	✓	✗	Reassignable variables
`const`	Block	✗	✗	✗	Fixed values

```
var name = "Ahmar";   // Function-scoped, can be changed or redeclared

let age = 23;         // Block-scoped, can change but not redeclare

const course = "AWP";  // Block-scoped, cannot change
```

Arrays

Arrays store multiple values in one variable.

Example:

```
const fruits = ["Apple", "Banana", "Mango"];
console.log(fruits[0]); // Output: Apple
```

You can loop through arrays using `for`, `forEach`, or array methods like `map()`.

Objects

Objects store **data in key–value pairs**.
They are useful for representing real-world entities (like students, users, or products).

Example:

```
const student = {
  name: "Ahmar",
  age: 23,
  course: "Advance Web Programming"
};

console.log(student.name); // Output: Ahmar
```

Functions

Functions allow you to **reuse code** and perform tasks.

Normal Function:

```
function greet(name) {
  return `Hello, ${name}!`;
}
```

Arrow Function:

```
const greet = (name) => `Hello, ${name}!`;
```

Difference:

- Arrow functions are shorter.
- They do not have their own `this` context (useful in React and callbacks).

Array Methods: map(), filter(), reduce()

map() : Used to transform each element of an array and return a new array.

```
const numbers = [1, 2, 3];
const doubled = numbers.map(num => num * 2);
console.log(doubled); // [2, 4, 6]
```

filter(): Used to select elements that meet a condition.

```
const numbers = [1, 2, 3, 4];
const even = numbers.filter(num => num % 2 === 0);
console.log(even); // [2, 4]
```

reduce() : Used to combine all elements into a single value (like sum or product).

```
const numbers = [1, 2, 3, 4];
const sum = numbers.reduce((acc, num) => acc + num, 0);
console.log(sum); // 10
```

Spread and Rest Operators

Spread Operator (. . .) : Expands array or object elements.

```
const arr1 = [1, 2];
const arr2 = [3, 4];
const merged = [...arr1, ...arr2];
console.log(merged); // [1, 2, 3, 4]
```

Rest Operator (. . .): Collects multiple elements into an array.

```
function sum(...numbers) {
  return numbers.reduce((a, b) => a + b);
}
console.log(sum(1, 2, 3, 4)); // 10
```

Quick Summary

Concept	Purpose	Example
HTML	Defines content	`<h1>Hello</h1>`
CSS	Styles the page	`h1 { color: blue; }`
JavaScript	Adds interactivity	`alert('Hi!')`
`let` & `const`	Modern variable declaration	`let x = 10;`
`map()`	Transform data	`[1,2,3].map(x=>x*2)`
`filter()`	Filter data	`[1,2,3].filter(x=>x>1)`
`reduce()`	Combine data	`[1,2,3].reduce((a,b)=>a+b)`
`...spread`	Expand elements	`[...arr1, ...arr2]`
`...rest`	Gather parameters	`function sum(...n){}`